Narratives, Noise Traders, and Asset Prices:

A General Equilibrium Approach

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Outline

1. Introduction

2. The Model

3. Analytical Results

Noise Traders

- Noise traders are broadly described as investors with trading behavior deviating from certain "benchmarks"

Category	Benchmark	Deviation Margin	Explanation
Belief Distortions	Full Information Rational Expectations (FI-RE)	Information Structure	Agents have distorted expectations about fundamentals.
Frictions	Frictionless Optimization	Feasible Set	Rational agents with complete information, but portfolio demand is distorted by market frictions.
Preference-Based	Standard Expected Utility	Preferences	Agents are fully informed and optimizing, but preferences depart from EU
Noise	Demand from Optimization	Source of Demand	Part of asset demand is an exogenous shock, unrelated to beliefs, preferences or frictions

- In this paper, noise traders are investors whose portfolio demand departs from the FI-RE benchmark
  - due to **Belief Distortions**

Narratives as Belief Distortions

- We focus on **Narratives** (Shiller, 2017, 2019),
  - a distorted belief about firms' prospects that spreads among investors
  - could be persistent or fleeting, broad or narrow
  - backed by no new information about fundamentals
- Real News vs. Narratives
  - Real news: new information about cash flows – changes what the firm is worth
  - Narratives: changes only noise trader beliefs
- Only noise traders trade on the misperception narratives create
  - markets clear, prices move – then drift back as the story fades
  - rational traders earn excess returns at noise traders' expense – a Sharpe ratio
  - Nothing fundamental changes – ***So, Do Narratives Matter?***

The Usual Approaches

1. **Theory.** Part of the system is held fixed (partial equilibrium) or greatly simplified
   - The right choice for isolating a mechanism – clearly showing a channel exists and how it works
   - Not appropriate for full picture – how the pieces interact, how effects are dampened or amplified
2. **Empirics.** Look for noise traders' effects in returns (APT factors, anomalies)
   - The correct discipline: beliefs can't be observed, effects must show up in data
   - Returns are where the field looks – if it doesn't show in average returns, it doesn't matter

- **This Paper**. GE with a genuine cross-section – many firms, a factor structure, and narratives. Then,
  - We derive the full mechanism, from beliefs to prices to wealth and back
  - We know where to look, and what to measure

What we Do

- J **firms and 2 fundamental risks** (2 Lucas trees). The J firms differ in
  1. Their exposure to the 2 trees
  2. The share of the trees' cash flows. The shares move idiosyncratically (martingale process)
- **Perpetual youth mode**l with two investor types, noise traders and rational traders
  - identical in preferences
  - different in beliefs: noise traders believe **narratives**, rational traders don't
- **Endogenous wealth** driven by two opposing forces:
  1. Noise traders gain / lose wealth with their trades
  2. New noise traders with endowments keep entering
  - the feedback between mispricing and noise-trader wealth is closed
- **Solved** by exploiting firm-level narratives are fleeting; **calibrated** to retail trading, holdings and news data

What we Find

1. **Narratives cause mispricing and transfer wealth** (from noise traders to rational traders)
   - many narratives, each diversifiable
   - mispricing through cross-section, systematic effects on prices and wealth
2. **The GE feedback loop caps mispricing effects**
   - Intense narratives → big mistakes → high Sharpe ratios → noise trader wealth depleted
   - so while they matter for prices and wealth, narratives leave little trace in average returns
3. **Frictions relaxes the cap**
   - heterogeneous risk aversion, passive capital, tracking-error limits, non-Merton demand
   - big mistakes go unarbitraged: detectable as factors with certain empirical methods

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Production

- **2 Lucas Trees** producing cash-flows D_k

$$\frac{\mathrm{d}D_k(t)}{D_k(t)} = \mu_{Dk}(\zeta)\,\mathrm{d}t + \sigma_{Dk}\,\mathrm{d}Z_{fk}(t), \qquad k = 1, 2$$

- **Mean-Reverting Shares.** $\mu_{Dk}(\zeta) = \bar{\mu} \mp \frac{\kappa_D}{2}(\zeta - \bar{\zeta})$; growth rates depending on $\zeta \equiv \ln(D_1/D_2)$
  - No tree dominates: whichever is smaller grows faster.
- J **Firms**, each owning a mix of the two trees, each representing a distinct segment in the economy:

	Tree 1	**Tree 2**
Projects	New ventures	Established operations
Style	Growth	Value

  - δ_j **(the share of firm j's value coming from tree 1)** places each firm on a spectrum between the two trees

Asset Menu

- Investors trade $J + 1$ assets
  1. A riskless bond in zero net supply, paying the rate $r(X, t)$ **(set in equilibrium)**
  2. J stocks, one per firm, each in unit supply
- For each stock $j \in \{1, \ldots, J\}$
  1. **Payout** (Π_j)**:** a share $\Upsilon_j \, s_{kj}$ of each tree's cash flows D_{kj}
     - $s_{kj} \in [0, 1]$ are the baseline shares, constant
     - Υ_j is an idiosyncratic multiplier, mean-reverting and driven by idiosyncratic Brownian $Z_{\iota j}$
  2. **Returns** (R_j)**:**

$$\mathrm{d}R_j \equiv \frac{\mathrm{d}P_j + \Pi_j \mathrm{d}t}{P_j} = \mu_{Rj}(X, t)\,\mathrm{d}t + \boldsymbol{\sigma}_{Rj}^{\top}(X, t) \underbrace{\mathrm{d}\mathbf{Z}_f}_{\text{Systematic}} + \sigma_{\iota j} \underbrace{\mathrm{d}Z_{\iota j}}_{\text{Idiosyncratic}}$$

Demographics

1. **Two Agent Types.** A fixed fraction $\alpha \in (0, 1)$ of Noise Traders, and $1 - \alpha$ Rational Traders
2. **Perpetual Youth.** **(Yaari 1965, Blanchard 1985)**
   - Continuous-time OLG with unit mass
   - Poisson exit rate χ with matched entry flow; fixed entry composition $(\alpha, 1 - \alpha)$, preserving demographic fractions
   - No bequest, actuarially fair annuities: active agents earn $r + \chi$, the problem stays scale invariant.
3. **Entry Endowment** $W_e = \tau Y / \chi$**.** Funded by a flat tax τ on output

Narratives

- **Narratives** distort what noise traders expect firms to earn by

$$b_{\ell j}\,\xi_\ell \quad \text{per year, for } j \in \mathcal{J}_\ell \subseteq \{1,\ldots,J\}$$

  - ξ_ℓ is intensity (either positive or negative), $b_{\ell j}$ is how much the narrative applies to firm j.

- **Dynamics.** Graph

$$\mathrm{d}\xi_\ell(t) = -\frac{1}{\varepsilon_\ell}\,\xi_\ell(t)\,\mathrm{d}t + \psi_\ell(t)\,\mathrm{d}N_\ell(t)$$

  - Spawns randomly at time $\tau_{\ell 1}$; decays with persistence $\varepsilon_\ell > 0$.
  - For $t > \tau_{\ell 1}$, each instant there is a possibility of a "flare-up" – a jump in intensity by $\psi_\ell(t)$
  - The chance of a flare-up rises with "popular interest"(fades over time) and "committed holdings":

$$\Pr(\mathrm{d}N_\ell(t) = 1) = \nu_\ell(t)\,\mathrm{d}t, \qquad \nu_\ell(t) = \nu_0 \sum_{m=1}^{N_\ell(t)} \exp\left(-\frac{t-\tau_{\ell m}}{T_A}\right) + \eta\,\|\mathbf{H}_\ell(t)\|_1$$

Optimization

- Wealth accumulates objectively (common for all investors)

$$\frac{\mathrm{d}W}{W} = \left[\boldsymbol{\pi}^\top \boldsymbol{\mu}_e + (r + \chi) - c\right] \mathrm{d}t + \boldsymbol{\pi}^\top \left(\Sigma_R \mathrm{d}\mathbf{Z} + \Sigma_\iota \mathrm{d}\mathbf{Z}_\iota + \Sigma_\xi \, \mathrm{d}\mathbf{Z}_\xi\right)$$

  - $\boldsymbol{\pi}$ is the vector of portfolio weights; $c \equiv C/W$ is consumption-wealth ratio

- The budget constraint used for optimizing:
  1. Rational Agents use objective dynamics
  2. Noise Traders use their own perceived returns, distorted by narratives ($\boldsymbol{\mu}_e \to \boldsymbol{\mu}_e^{\mathcal{N}}$)

- Type specific **Optimal Policies,** (derived from ▸ type specific HJB Equations) for $i \in \{\mathcal{N}, \mathcal{R}\}$,

$$\boxed{c^{i*} = v^i(\hat{X}, t)^{-1/\gamma}, \qquad \boldsymbol{\pi}^{i*} = \frac{1}{\gamma}\hat{\Omega}^{-1}\boldsymbol{\mu}_e^i + \frac{1}{\gamma}\hat{\Omega}^{-1}\Sigma_{R\hat{X}}\nabla \ln v^i(\hat{X}, t)}$$

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Some Notation

1. Type specific wealth $W^{\mathcal{N}}, W^{\mathcal{R}}$ and the noise trader wealth share $\omega \equiv W^{\mathcal{N}}/(W^{\mathcal{N}} + W^{\mathcal{R}})$
2. Wealth weighted averages: $\bar{y} = \omega\, y^{\mathcal{N}} + (1-\omega)\, y^{\mathcal{R}}$
   - *(e.g.)* $\bar{\boldsymbol{\pi}} = \omega\, \boldsymbol{\pi}^{\mathcal{N}*} + (1-\omega)\, \boldsymbol{\pi}^{\mathcal{R}*}, \qquad \bar{c} = \omega\, c^{\mathcal{N}*} + (1-\omega)\, c^{\mathcal{R}*}$
3. Disagreement: $\Delta y = y^{\mathcal{N}} - y^{\mathcal{R}}$
   - *(e.g.)* $\Delta\boldsymbol{\pi} = \boldsymbol{\pi}^{\mathcal{N}*} - \boldsymbol{\pi}^{\mathcal{R}*}, \qquad \Delta c = c^{\mathcal{N}*} - c^{\mathcal{R}*}$
4. Because (1) the risk-free asset is in zero net supply, and (2) the risky asset markets clear:

$$\underbrace{\bar{\pi}_j}_{\text{Aggregate Demand}} = \underbrace{\frac{P_j}{P}}_{\text{Market Portfolio}}, \qquad j = 1, \ldots, J$$

   - $P = \sum_j P_j$ is the total market value.

Workhorse Equation (1): Equilibrium Price Premium

1. **Valuation Premium**: $\boxed{\tilde{\varepsilon}(t)\, b_{\ell j}\, \xi_\ell(t)}$ **(PV of all extra returns promised on firm j by narrative ℓ; only perceived by noise traders)**
   - $\tilde{\varepsilon} \equiv \frac{\varepsilon}{1+r(X,t)\,\varepsilon}$ **(effective years the narrative has left; same across all narratives in baseline)**

2. Markets Clear:

$$\underbrace{\frac{\omega}{\gamma_{\mathcal{N}}}\left(\tilde{\varepsilon}\, b_{\ell j}\xi_\ell - p_j\right)}_{\textit{Willingness to Buy}} = \underbrace{\frac{(1-\omega)}{\gamma_{\mathcal{R}}}\left(p_j - 0\right)}_{\textit{Willingness to Sell}} \implies \boxed{p_j = \varpi\, \tilde{\varepsilon}\, b_{\ell j}\, \xi_\ell}$$

Equilibrium Price Premium

- $p_j \equiv \ln\left(\frac{P_j}{P_j^*}\right)$, and P_j^* is the price of claim j when there are no narratives
- $\varpi \equiv \frac{\omega/\gamma_{\mathcal{N}}}{\omega/\gamma_{\mathcal{N}}+(1-\omega)/\gamma_{\mathcal{R}}}$ measures noise traders' share of total risk-bearing capacity.

  ($\star$ In the baseline case with $\gamma_{\mathcal{N}} = \gamma_{\mathcal{R}}$, boils down to $\varpi = \omega$)

Workhorse Equation (2): Market Selection

1. **Equilibrium Expected Excess Returns** ($\boldsymbol{\mu}_e$). Since p_j fades as narratives fade ($\mathrm{d}p_j = -\frac{p_j}{\varepsilon}\mathrm{d}t$),

$$\mathrm{d}p_j - r\,p_j\mathrm{d}t = -\frac{p_j}{\tilde{\varepsilon}}\,\mathrm{d}t = \underbrace{-\,\varpi\, b_{\ell j}\,\xi_\ell}_{\text{Belief Discount}}\,\mathrm{d}t \quad \Longrightarrow \quad \boldsymbol{\mu}_e = \boldsymbol{\mu}_e^* - \varpi\,\mathbf{B}\boldsymbol{\xi}$$

   - $\boldsymbol{\mu}_e^*$ is excess returns without narratives, $\mathbf{B} = [\mathbf{b}_1 \cdots \mathbf{b}_L]$, and $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_L)^\top$.

2. **Market Selection:** noise traders' wealth share moves with their mistakes

$$\mathrm{d}\omega = \underbrace{\omega(1-\omega)\left[(\Delta\boldsymbol{\pi})^\top(\boldsymbol{\mu}_e - \hat{\Omega}\bar{\boldsymbol{\pi}}) - \Delta c\right]}_{\text{market selection}}\mathrm{d}t + \underbrace{\frac{\tau}{q}(\alpha-\omega)}_{\text{replenishment}}\,\mathrm{d}t + (\text{diffusion})$$

   - ★ In the baseline case ($\gamma_{\mathcal{N}} = \gamma_{\mathcal{R}} = \gamma$): $\Delta\boldsymbol{\pi} = \frac{1}{\gamma}\hat{\Omega}^{-1}\mathbf{B}\boldsymbol{\xi} \Longrightarrow (\Delta\boldsymbol{\pi})^\top(\boldsymbol{\mu}_e - \hat{\Omega}\bar{\boldsymbol{\pi}}) \leq 0$, for any $\boldsymbol{\xi}$

Frictionless Economy: Stationary Wealth

- Noise trader's survive at an interior wealth share ω^*
  - Losses from mistakes (narratives) add up in noise trader wealth vs. replenishment
  - The two opposing forces balance at:

$$\omega^*(1-\omega^*)\frac{\varpi^*}{\gamma}\overline{(\mathbf{B}\boldsymbol{\xi})^\top\hat{\Omega}^{-1}(\mathbf{B}\boldsymbol{\xi})} = \frac{\tau}{q}(\alpha-\omega^*)$$

- Since $\varpi\,\mathbf{B}\,\boldsymbol{\xi}$, is the belief discount, *i.e.* the excess rational traders can earn by leaning on narratives, the (averaged) narrative Sharpe is $\bar{\mathcal{S}}^2 = (\varpi^*)^2\overline{(\mathbf{B}\boldsymbol{\xi})^\top\hat{\Omega}^{-1}(\mathbf{B}\boldsymbol{\xi})}$. The balance equation above:

$$\frac{\omega^*(1-\omega^*)}{\gamma}\bar{\mathcal{S}}^2 = \frac{\tau}{q}(\alpha-\omega^*) \quad\Longrightarrow\quad \boxed{\bar{\mathcal{S}}^2 = \gamma\frac{\varpi^*}{\omega^*}\frac{\frac{\tau}{q}(\alpha-\omega^*)}{1-\omega^*}}$$

Thank You

Thank You

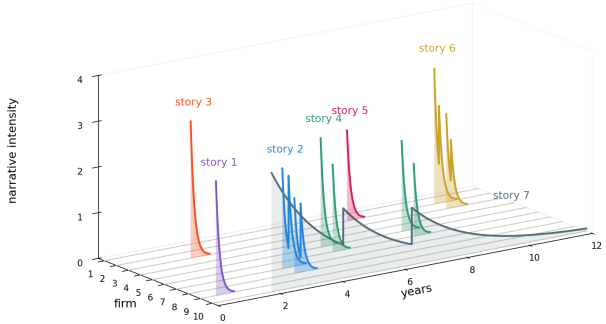
Data Work

return

- **Data Used**: TAQ (2010-2024), CRSP, RavenPack, Compustat, Robintrack
- **Identification Strategy**
  1. Step 1. Define Related Concepts (all definitions & thresholds are fixed beforehand)
     - *Retail Surges*: days when a stock's net retail buying/selling is unusually large
     - *Real News*: earnings announcements / new & relevant Dow Jones story **(from Compustat, RavenPack)**
     - *Narratives*: no news, but unusually heavy web coverage **(from RavenPack Web Edition)**
  2. Step 2. Classify each retail surge by what surrounds it
     - *News-Driven*: retail surges with real relevant news within two days
     - *Narrative-Driven* ***(flare-ups)***: retail surges with no news but heavy web coverage observed
     - *Unexplained*: neither
  3. Step 3. Separate stories from reaction to prices,
  4. Step 4. Compare narrative-driven with control group (news-driven) $\longrightarrow$ isolate effect of narrative

Narratives: Dynamics

return



Hamilton-Jacobi-Bellman Equation

Solving Type-i Agent's Problem: $V^i(W, X, \boldsymbol{\xi}, t) = \max_{\{c,\boldsymbol{\pi}\}} \mathbb{E}_t^i \left[\int_t^\infty e^{-(\rho+\chi)(u-t)} \frac{(cW)^{1-\gamma}}{1-\gamma} \, \mathrm{d}u \right]$

- HJB (with $\hat{\Omega}$ the total covariance of returns and $\Sigma_{R\hat{X}}$ their covariance with the state)

$$
\begin{aligned}
0 = \max_{c,\boldsymbol{\pi}} \Bigg\{ & \frac{(cW)^{1-\gamma}}{1-\gamma} - (\rho+\chi)V^i + [\boldsymbol{\pi}^\top \boldsymbol{\mu}_e^i + (r+\chi) - c] W \partial_W V^i \\
& + \frac{1}{2} \boldsymbol{\pi}^\top \hat{\Omega} \boldsymbol{\pi} W^2 \partial_{WW} V^i + \mathcal{L}_{\hat{X}}^i V^i + W \boldsymbol{\pi}^\top \Sigma_{R\hat{X}} \partial_{W\hat{X}} V^i \Bigg\},
\end{aligned}
$$

- Reduced HJB with $V^i = \left(W^{1-\gamma}/(1-\gamma)\right) v^i$

$$
\frac{\mathcal{L}_{\hat{X}}^i v^i}{v^i} + c^i - (\rho+\chi) + (1-\gamma) \left[(\boldsymbol{\pi}^i)^\top \boldsymbol{\mu}_e^i + r + \chi - c^i - \frac{\gamma}{2} (\boldsymbol{\pi}^i)^\top \hat{\Omega} \boldsymbol{\pi}^i + (\boldsymbol{\pi}^i)^\top \Sigma_{R,\hat{X}} \nabla \ln v^i \right] = 0
$$

return